

Dark Energy as Residual Vacuum Free Energy: A Thermodynamic Bound on the Cosmological Constant

Part of the FST Program [35]

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Abstract

We study a thermodynamic framework in which dark energy is associated with residual vacuum free energy. For the dimensionally closed reduced function $\Phi_{\text{red}} = A + B\rho \ln(\rho/\rho_{\text{Pl}}) + C\rho^2$, $B, C > 0$, we prove existence, uniqueness, strict convexity, and an explicit Lambert- W minimizer. The original momentum-space expression is dimensionally inhomogeneous and does not determine the matching coefficients. The scaling $\Lambda_{\text{eff}} \sim H_0^2/[c^2 \ln(M_{\text{Pl}}/H_0)]$ therefore remains an external RG-matching ansatz. It gives $3.8 \times 10^{-55} \text{ m}^{-2}$, about 290 times below observation.

The Pöschl–Teller potential $V_0 \text{sech}^2(\phi/\phi_0)$ has an unstable maximum at $\phi = 0$; a tanh history is a phenomenological proxy rather than a derived stable attractor. Standard quasistatic metric $f(R)$ perturbations approach a slip of $1/2$ (or reciprocal 2, by convention) and a GR-like lensing combination $\Sigma \simeq 1/F$. The Hu–Sawicki environment ledger supplies necessary curvature thresholds, but a source-bound local profile is still missing; neither $|f_{R0}| = 1.2 \times 10^{-5}$ nor 10^{-6} alone establishes Cassini compatibility. The current result is a proved reduced mathematical model within an ongoing physical research programme. RG matching, a stable scalar history, local screening, and official cosmological likelihood tests remain open.

Keywords: cosmological constant, dark energy, vacuum energy, scalar-tensor gravity, gravitational slip

Contents

1	Introduction	3
2	Dimensionally Closed Variational Model	4
2.1	Raw Motivation and Its Limitations	4
2.2	Reduced Model	4
3	Proven Result and Conditional Scaling	5
3.1	Scalar-Sector Status	6
4	Kinematic Consequences of a Density Proxy	7
5	Local Consistency Gates	7
5.1	Radiation-Era Trace	7
5.2	Convexity and Screening	7
5.3	Scope of the Variational Statement	8
6	Renormalization, Beta Response, and Observational Status	8
6.1	Metric $f(R)$ Perturbations	9
6.2	Source-Bound Local Gate	9
7	Proof Status and Next Steps	10

1 Introduction

This paper develops a thermodynamic research framework for dark energy and states its current mathematical and physical status. The reduced variational model has a proven unique minimum; connecting that minimum to a gravitating cosmological density remains an open matching problem.

Four established limitations guide the continuing programme. The original momentum-space integrand is dimensionally inhomogeneous and does not define a physical free-energy density without explicit matching coefficients. The Pöschl–Teller hilltop is unstable, and the proposed saturation law does not follow from the canonical Klein–Gordon–Friedmann system. A thermodynamic beta function defined by differentiating a stationarity identity is identically zero. Standard metric $f(R)$ perturbations give limits different from the earlier forecasts $\eta = 5/4$ and $\Sigma = 9/8$. Those forecasts and the stable-attractor interpretation are not supported; their history and the underlying derivations are documented below. Table 1 records the current evidence and remaining requirements, while Figure 1 summarizes the four research layers.

Table 1: Current evidence and remaining requirements for the research programme.

Layer	What is established here	Stop gate before a stronger claim
Reduced variational core	Existence, uniqueness, strict convexity, and the Lambert- W minimizer for specified $B, C > 0$.	Independent physical derivation of $B(H)$, $C(H)$, and the subtraction scheme.
Conditional cosmological scale	The corrected ansatz value is $3.8 \times 10^{-55} \text{ m}^{-2}$, about 290 below observation.	No target-value input when fixing C/B ; compare only after the RG ledger is closed.
Scalar and perturbation sector	Earlier hilltop, beta-response, $\eta = 5/4$, and $\Sigma = 9/8$ claims are withdrawn.	Stable scalar history and standard $f(R)$ response must be derived from the full equations.
Local and observational gates	Environment thresholds are arithmetically defined; proxy MCMC is background-level only.	Source-bound local $f_R(\text{env})$ profile and official Planck/DESI likelihood.
RVM/entropy audit	The 3 August 2026 evaluator is only an exploratory insertion calculation.	Derive ν , horizon entropy, $A(E)$, ensemble, and Cassini convention without inserted pass values.

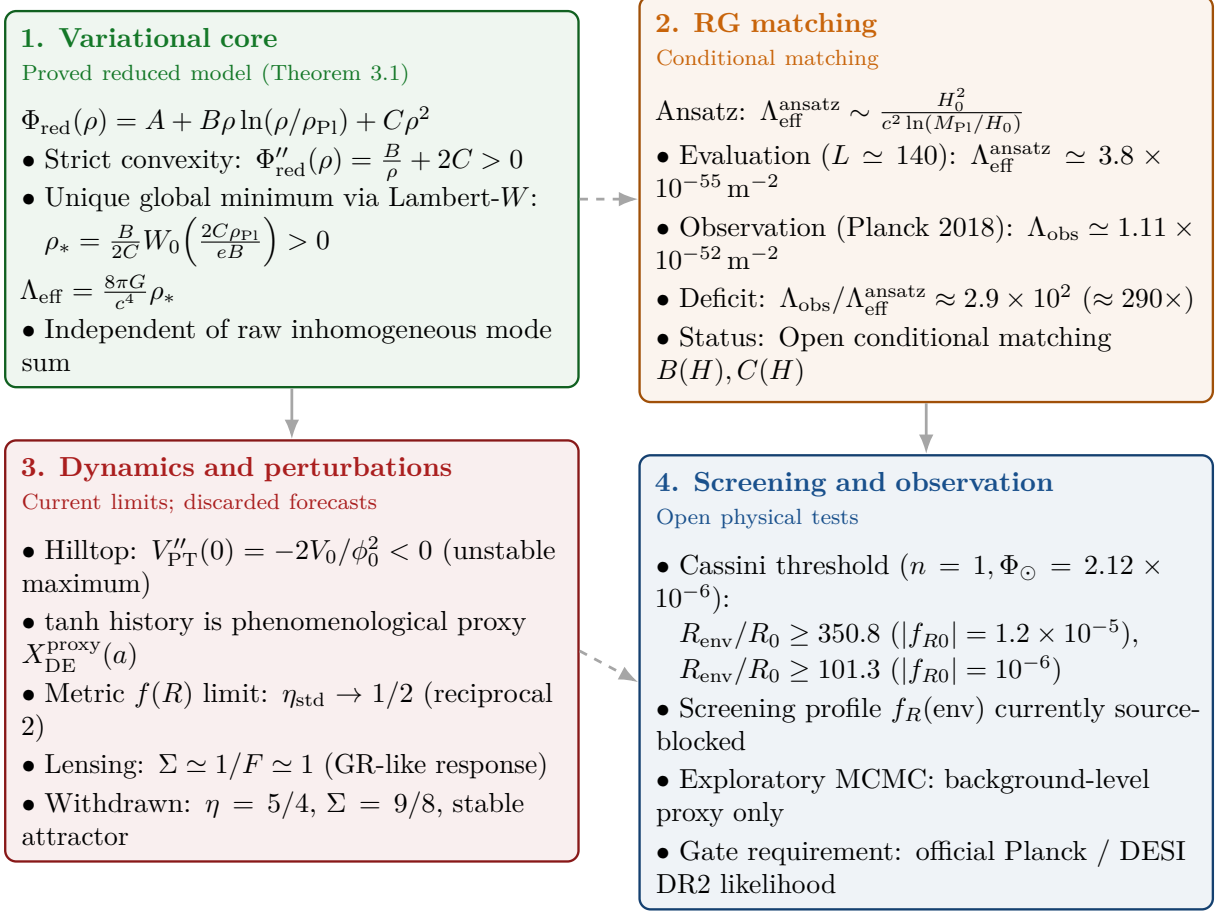


Figure 1: Current research architecture of the FST dark-energy framework: the proved reduced variational model (green), conditional RG matching (orange), dynamical limits and discarded forecasts (red), and open local screening and likelihood tests (blue). Arrows mark connections and comparison steps that do not constitute completed physical derivations.

2 Dimensionally Closed Variational Model

2.1 Raw Motivation and Its Limitations

The earlier motivating expression was

$$\Phi_{\text{raw}}(\rho) = \int_{\Lambda_{\text{IR}}}^{\Lambda_{\text{UV}}} \left[\frac{\omega(k)}{2} + T_{\text{dS}} \rho \ln\left(\frac{\rho}{\rho_{\text{Pl}}}\right) + \frac{8\pi G \rho^2}{k^2} \right] \frac{4\pi k^2}{(2\pi)^3} dk. \quad (1)$$

In natural units, the three terms inside the square brackets have dimensions E , E^5 , and E^4 , respectively. They cannot be added as written. Planck-unit rescaling does not repair the problem: it hides powers of M_{Pl} and H , precisely the powers that determine the proposed cosmological scaling. Equation (1) is therefore only a schematic motivation, not a physical functional and not a derivation from a mode sum.

2.2 Reduced Model

Definition 2.1 (Reduced Free-Energy Density). For $\rho > 0$, let

$$\Phi_{\text{red}}(\rho) = A + B \rho \ln\left(\frac{\rho}{\rho_{\text{Pl}}}\right) + C \rho^2, \quad B > 0, \quad C > 0. \quad (2)$$

Here $[\rho] = [\Phi_{\text{red}}] = E^4$, $[A] = E^4$, $[B] = 1$, and $[C] = E^{-4}$. The constants B and C are renormalized matching coefficients. Their values are not supplied by Eq. (1).

The kernel

$$V_{\text{grav}}(\rho, k) = \frac{8\pi G \rho^2}{k^2} \quad (3)$$

has the dimension of an energy density and can motivate a positive quadratic term after an explicitly specified integration and matching prescription. Dimensional analysis selects it as a lowest-order monomial only after the assumptions of one power of G , quadratic dependence on ρ , and inverse powers of k are imposed. It does not establish uniqueness among general kernels, and a quasistatic 4/3 force enhancement does not by itself derive an off-shell free-energy functional.

3 Proven Result and Conditional Scaling

Theorem 3.1 (Unique Minimizer of the Reduced Model). *The function Φ_{red} in Definition 2.1 has exactly one global minimizer $\rho_* > 0$. It satisfies*

$$B \left[1 + \ln \left(\frac{\rho_*}{\rho_{\text{Pl}}} \right) \right] + 2C\rho_* = 0 \quad (4)$$

and is given explicitly by

$$\rho_* = \frac{B}{2C} W \left(\frac{2C\rho_{\text{Pl}}}{eB} \right), \quad (5)$$

where W is the principal Lambert function.

Proof. The first derivative tends to $-\infty$ as $\rho \rightarrow 0^+$ and to $+\infty$ as $\rho \rightarrow \infty$. Moreover,

$$\Phi_{\text{red}}''(\rho) = \frac{B}{\rho} + 2C > 0. \quad (6)$$

Thus the derivative has exactly one zero and strict convexity makes that zero the unique global minimizer. Rearranging Eq. (4) gives $(2C\rho_*/B) \exp(2C\rho_*/B) = 2C\rho_{\text{Pl}}/(eB)$, which yields Eq. (5). $\square$

The associated curvature convention is

$$\Lambda_{\text{eff}} = \frac{8\pi G}{c^4} \rho_*. \quad (7)$$

No dependence on H_0 follows until B/C is independently matched. The frequently used expression

$$\Lambda_{\text{eff}}^{\text{ansatz}} \sim \frac{H_0^2}{c^2 \ln(M_{\text{Pl}}/H_0)} \quad (8)$$

is therefore a conditional RG-matching ansatz, not a consequence of Theorem 3.1.

Corollary 3.2 (Arithmetic of the Conditional Ansatz). *For $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $L = \ln(M_{\text{Pl}}/H_0) \simeq 140$,*

$$\frac{H_0^2}{c^2 L} \simeq 3.8 \times 10^{-55} \text{ m}^{-2}. \quad (9)$$

Compared with $\Lambda_{\text{obs}} \simeq 1.11 \times 10^{-52} \text{ m}^{-2}$ from Planck 2018 [4], the ansatz is low by a factor of about 2.9×10^2 . This is an unresolved normalization/matching factor, not order-of-magnitude agreement.

Physical-identification gate. The curvature convention in Eq. (7) is conditional on an independent identification of the model parameter with the gravitating energy density, including its conservation or exchange law and the subtraction term A . No external RVM or horizon result fixes these choices or removes the factor-290 matching deficit automatically.

3.1 Scalar-Sector Status

Consider the phenomenological Lagrangian

$$\mathcal{L} = \frac{R + \gamma R^2}{16\pi G} + \frac{1}{2}(\partial\phi)^2 - V_{\text{PT}}(\phi) + \mathcal{L}_m, \quad V_{\text{PT}}(\phi) = V_0 \text{sech}^2(\phi/\phi_0). \quad (10)$$

For the quadratic curvature sector the scalaron mass is $m_s^2 = 1/(6\gamma)$. The scalar potential obeys

$$V'_{\text{PT}}(0) = 0, \quad V''_{\text{PT}}(0) = -\frac{2V_0}{\phi_0^2} < 0. \quad (11)$$

Hence $\phi = 0$ is a maximum, not a minimum. Linear perturbations satisfy

$$r^2 + 3Hr - \frac{2V_0}{\phi_0^2} = 0,$$

and the larger root is positive for every $V_0 > 0$. Hubble friction can delay the roll but cannot make the hilltop asymptotically stable. The earlier stable de Sitter theorem and oscillation forecast are withdrawn.

For a canonical scalar and pressureless matter, define

$$x = \frac{\dot{\phi}}{\sqrt{6} H M_{\text{Pl}}}, \quad y = \frac{\sqrt{V}}{\sqrt{3} H M_{\text{Pl}}}, \quad \lambda = -M_{\text{Pl}} \frac{V'}{V}, \quad N = \ln a.$$

The correct autonomous system is

$$x' = -3x + \sqrt{\frac{3}{2}} \lambda y^2 + \frac{3}{2} x(1 + x^2 - y^2), \quad (12)$$

$$y' = -\sqrt{\frac{3}{2}} \lambda x y + \frac{3}{2} y(1 + x^2 - y^2). \quad (13)$$

With $\Omega_\phi = x^2 + y^2$ and $w_\phi = (x^2 - y^2)/(x^2 + y^2)$,

$$\frac{d\Omega_\phi}{dN} = -3w_\phi \Omega_\phi (1 - \Omega_\phi). \quad (14)$$

This does not reduce to the tanh equation printed in earlier versions.

A bounded tanh curve may still be used as a data parameterization if it is labeled as such. A dimensionally consistent non-negative choice is

$$X_{\text{DE}}^{\text{proxy}}(a) = \frac{\Phi_0}{2} \left[1 + \tanh(\kappa(a - a_t)) \right], \quad (15)$$

$$\frac{dX_{\text{DE}}^{\text{proxy}}}{da} = 2\kappa X_{\text{DE}}^{\text{proxy}} \left(1 - \frac{X_{\text{DE}}^{\text{proxy}}}{\Phi_0} \right). \quad (16)$$

Here $X_{\text{DE}} = \rho_{\text{DE}}/\rho_{\text{crit},0}$, not the instantaneous fraction $\rho_{\text{DE}}/\rho_{\text{crit}}(a)$. Equations (15)–(16) are a phenomenological proxy and are not derived from Eq. (10).

4 Kinematic Consequences of a Density Proxy

The conditional estimate from Eq. (8) is $3.8 \times 10^{-55} \text{ m}^{-2}$, not $3.8 \times 10^{-53} \text{ m}^{-2}$. It is therefore about 290 times below Λ_{obs} , not a factor of three.

A bounded curve may be used as a phenomenological density history only if its normalization is defined consistently. Let $X_{\text{DE}}(a) = \rho_{\text{DE}}(a)/\rho_{\text{crit},0}$ and

$$E^2(a) = \frac{H^2}{H_0^2} = \Omega_m a^{-3} + X_{\text{DE}}(a).$$

Then

$$q(a) = -1 - \frac{a}{2E^2} \frac{dE^2}{da} = \frac{\frac{1}{2}\Omega_m a^{-3} - X_{\text{DE}} - \frac{a}{2}X'_{\text{DE}}}{E^2}, \quad (17)$$

$$w_{\text{DE}}(a) = -1 - \frac{1}{3} \frac{d \ln X_{\text{DE}}}{d \ln a}. \quad (18)$$

The earlier deceleration formula used the wrong coefficient for aX'_{DE} . It also alternated between an instantaneous density fraction and a density normalized to today's critical density. Consequently, the reported acceleration redshift, w_0 , CPL map, and proxy χ^2 are withdrawn pending recomputation from a physically derived history. No official DESI likelihood evaluation is claimed.

Remark 4.1 (DESI DR2 Comparators and the Open Model-Identification Gate). Scalar-tensor reconstructions [36] and interacting quintessence with field-dependent dark-matter masses [37] provide distinct mechanisms for an effective phantom crossing. The quintom model of [38] instead includes a canonical and a phantom field; it is not the same ghost-free interacting-quintessence mechanism. These external models do not identify the reduced minimizer with a conserved gravitational energy density or close its matching, exchange, horizon/ensemble, local-screening, or likelihood gates. The tanh proxy alone establishes no phantom-crossing mechanism.

5 Local Consistency Gates

5.1 Radiation-Era Trace

For an ideal perfect fluid with the sign convention used here,

$$T = (1 - 3w)\rho.$$

Thus $T = 0$ for exactly massless radiation with $w = 1/3$. This removes the leading classical trace source, but it is not a proof of BBN safety: massive species, trace anomalies, background evolution, and initial conditions must be propagated through a quantitative abundance calculation.

5.2 Convexity and Screening

A fixed strictly convex function of ρ has one minimizer. This elementary fact does not constitute a no-go theorem for chameleon screening. In a chameleon model the effective scalar potential depends on ambient density, so the function being minimized changes with the environment. The previous Γ -convergence and energy-dissipation block did not supply

a valid liminf/recovery proof and mixed gradients of different variables; it is withdrawn. The accompanying numerical phase plot remains, at most, a toy illustration and not a Cassini calculation.

5.3 Scope of the Variational Statement

The variational theorem below concerns only the reduced scalar function with specified positive coefficients. It does not solve the radiative-stability problem, determine a vacuum subtraction scheme, construct an RG trajectory, or derive a local screening profile. Those are independent gates. In particular, neither a resolvent analogy nor a metric on configuration space changes the zero set of the differential of a fixed scalar functional.

6 Renormalization, Beta Response, and Observational Status

The physical scaling problem is concentrated in the matching coefficients B and C . A valid derivation must provide a source- and scheme-controlled ledger

bare $\longrightarrow$ Minkowski subtraction $\longrightarrow$ counterterms $\longrightarrow$ horizon remainder.

No such derivation is supplied here. The RG, data-processing, one-loop, BBN, and asymptotic-safety passages in the earlier skeleton were analogies or leading-order observations, not proofs. In particular, the ideal-radiation identity $T = (1 - 3w)\rho = 0$ is correct for $w = 1/3$, but it does not by itself guarantee sub-percent BBN safety once massive species, trace anomalies, and the background scalar dynamics are included.

The earlier definition

$$\beta_{\Lambda}^{\text{old}} = \frac{d}{d \ln a} \left. \frac{\partial \Phi}{\partial \rho} \right|_{\rho=\rho_*(a)}$$

is identically zero because the bracket is zero for every a . It cannot yield a non-zero flow. If one merely differentiates the conditional ansatz $\rho_*^{\text{ansatz}} \propto H^2 M_{\text{Pl}}^2 / L$, $L = \ln(M_{\text{Pl}}/H)$, the result is

$$\frac{d\rho_*^{\text{ansatz}}}{d \ln a} = \rho_*^{\text{ansatz}} \left(2 + \frac{1}{L} \right) \frac{d \ln H}{d \ln a}. \quad (19)$$

This is a response of an assumed scaling, not an independently extracted beta function and not evidence for that scaling.

No Wilsonian beta function, fixed-point monotonicity, bound on γ at one loop, or Kullback–Leibler data-processing statement is promoted to a result in the present framework. Each requires a dimensionally complete action, normalized probability objects where information inequalities are invoked, and an explicit renormalization prescription.

The conditional scale

$$\Lambda_{\text{eff}}^{\text{ansatz}} \simeq 3.8 \times 10^{-55} \text{ m}^{-2}$$

is about 290 times below the observed value. Accordingly, this paper does not claim a solution of the cosmological-constant or coincidence problems, a DESI-compatible equation of state, or a stable late-time attractor.

6.1 Metric $f(R)$ Perturbations

Use the convention

$$ds^2 = -(1 + 2\Psi)dt^2 + a^2(1 - 2\Phi)d\mathbf{x}^2$$

and define $s(k) = (k^2/a^2)/(k^2/a^2 + m_s^2)$. For $F = df/dR \simeq 1$, the quasistatic linear response has the standard form

$$k^2\Psi = -4\pi G a^2 \bar{\rho} \delta \left(1 + \frac{s}{3}\right), \quad (20)$$

$$k^2\Phi = -4\pi G a^2 \bar{\rho} \delta \left(1 - \frac{s}{3}\right). \quad (21)$$

Thus, with the standard numerator convention,

$$\eta_{\text{std}} = \frac{\Phi}{\Psi} = \frac{1 - s/3}{1 + s/3} \longrightarrow \frac{1}{2} \quad (s \rightarrow 1), \quad (22)$$

while the reciprocal convention tends to 2. The Weyl/lensing combination remains GR-like, apart from the usual $1/F$ factor:

$$\Sigma \simeq \frac{1}{F} \simeq 1. \quad (23)$$

Consequently, $\eta = 5/4$ and $\Sigma = 9/8$ are not predictions of standard metric $f(R)$ gravity and all forecasts based on those values are withdrawn. The $4/3$ enhancement applies to the force potential Ψ , not to a universal free-energy coefficient [13, 14].

Strict convexity of a fixed, environment-independent scalar function $\Phi_{\text{red}}(\rho)$ excludes a second minimum of that same function. It does not rule out chameleon screening, because the effective scalar potential itself changes with ambient density. Likewise, changing only a positive-definite metric on configuration space cannot move the minimizer of a fixed scalar function: $\text{grad}_g \Phi = 0$ if and only if $d\Phi = 0$.

6.2 Source-Bound Local Gate

The 2 July 2026 environment ledger uses

$$|f_R(\text{env})| = |f_{R0}| \left(\frac{R_0}{R_{\text{env}}} \right)^{n+1} \quad (24)$$

together with

$$\epsilon_{\text{th}} = \frac{|f_{R,\text{env}} - f_{R,\text{in}}|}{2\Phi_{\odot}}, \quad f_{R,\text{in}} \simeq 0, \quad \Phi_{\odot} = 2.12 \times 10^{-6}.$$

The Cassini threshold $\epsilon_{\text{th}} < 2.3 \times 10^{-5}$ requires

$$\frac{|f_R(\text{env})|}{|f_{R0}|} < \frac{2\Phi_{\odot}(2.3 \times 10^{-5})}{|f_{R0}|}. \quad (25)$$

For $n = 1$, this gives

$$\frac{R_{\text{env}}}{R_0} \geq 350.787 \quad \text{for } |f_{R0}| = 1.2 \times 10^{-5}, \quad \frac{R_{\text{env}}}{R_0} \geq 101.264 \quad \text{for } |f_{R0}| = 10^{-6}.$$

These threshold calculations are arithmetically correct. No independent local density/curvature model currently supplies the required R_{env}/R_0 ; the ledger therefore permits no claim upgrade. Neither the exploratory best-fit 1.2×10^{-5} nor the conservative 10^{-6} guide is by itself a Cassini proof. The former is background-proxy compatible only and is not demonstrated Cassini-safe.

The previous exploratory MCMC scan used simplified diagonal summary terms rather than the official Planck/DESI likelihoods and encoded Cassini through a linear hard proxy. Its numerical samples may be retained as a background stress test only; they do not establish model preference, official parameter constraints, or local screening. Table 2 provides a synoptic synthesis of the quantitative parameters, relations, and empirical thresholds governing each sector.

Table 2: Synoptic Parameter, Bound, and Falsification Ledger Across All Sectors of the Framework.

Sector / Quantity	Formal Expression	Value / Bound	Reference / Gate	Epistemic Status
Variational minimizer	$\rho_* = \frac{B}{2C} W_0 \left(\frac{2C\rho_{\text{Pl}}}{eB} \right)$	$\rho_* > 0$ for $B, C > 0$	Def. 2.1, Eq. (5)	Proven (strictly convex ground state)
Curvature mapping	$\Lambda_{\text{eff}} = 8\pi G\rho_*/c^4$	Dependent on B/C	Eq. (7)	Exact mapping
Cosmological ansatz	$\Lambda_{\text{eff}}^{\text{ansatz}} \sim \frac{H_0^2}{c^2 \ln(M_{\text{Pl}}/H_0)}$	$3.8 \times 10^{-55} \text{ m}^{-2}$	Planck 2018 ($1.11 \times 10^{-52} \text{ m}^{-2}$)	Conditional (290× deficit; B, C open)
Pöschl–Teller curvature	$V_{\text{PT}}''(0) = -2V_0/\phi_0^2$	Negative ($r_+ > 0$)	Eq. (11)	Unstable hilltop (tanh proxy only)
Scalaron mass	$m_s^2 = 1/(6\gamma)$	$\gamma > 0$	Eq. (10)	Standard R^2 gravity
Gravitational slip	$\eta_{\text{std}} = \frac{1-s/3}{1+s/3} \rightarrow 1/2$	Reciprocal $\rightarrow 2$	Eq. (22)	Standard metric limit (5/4 withdrawn)
Lensing parameter	$\Sigma \simeq 1/F \simeq 1$	GR-like response	Eq. (23)	Standard metric limit (9/8 withdrawn)
Cassini gate (1.2×10^{-5})	$R_{\text{env}}/R_0 \geq 350.787$	$\epsilon_{\text{th}} < 2.3 \times 10^{-5}$	Eq. (24)	Arithmetic pass; local profile source-blocked
Cassini gate (10^{-6})	$R_{\text{env}}/R_0 \geq 101.264$	$\epsilon_{\text{th}} < 2.3 \times 10^{-5}$	Eq. (24)	Conservative guide; no screening proof
Radiation trace	$T = (1 - 3w)\rho$	$T = 0$ for $w = 1/3$	Massless fluid	Exact identity; BBN quantitative gate open

7 Proof Status and Next Steps

Table 3 records the current status of the seven research components, distinguishing proved mathematical results, conditional ansätze, and unsupported earlier assertions.

The next decisive calculations are: (i) determine dimensionally explicit, renormalized $B(H)$ and $C(H)$; (ii) replace the hilltop potential or derive a stable scalar history from the full equations; (iii) supply an independent galactic/solar Hu–Sawicki profile whose curvature ratio meets the ledger threshold; and (iv) only then rerun an official cosmological

Table 3: Current mathematical and physical status of the research programme.

Item	Current status
Reduced functional	Proven: existence, uniqueness, strict convexity, and the Lambert- W minimizer for specified $B, C > 0$.
Raw mode sum	Invalid as a dimensionally homogeneous functional; motivation only.
Physical $H_0^2/\ln$ scaling	Open and conditional on explicit RG matching; current arithmetic misses observation by about 290.
Pöschl–Teller dynamics	Hilltop transient; no stable de Sitter attractor and no derived tanh law.
Metric $f(R)$ slip/lensing	Standard unscreened limits are $\eta_{\text{std}} \rightarrow 1/2$ (or reciprocal 2) and $\Sigma \simeq 1/F$, not $5/4$ and $9/8$.
Local screening	Open: threshold ledger correct, but all numerical passes are source-blocked.
Exploratory MCMC	Background-level proxy only; no official Planck/DESI likelihood and no Cassini-safe classification.

likelihood. These gates block stronger physical claims and a validated-model release. The programme continues from the proved reduced mathematical result, with these open requirements defining the next physical tests.

Development since Version 1.12

- The abstract, introduction, and evidence tables now describe the current research programme directly; the title and scientific identity remain unchanged.
- Figure 1 gives headings, status lines, and content separate space. The corrected dynamics are distinguished from the specific discarded forecasts.
- The proved reduced variational model, the factor-290 matching deficit, the unstable hilltop, the standard $f(R)$ limits, and the open screening and likelihood requirements retain their previous evidentiary status.
- No additional physical evidence, computation, counterexample, or validation is introduced in this version. Earlier counterarguments and discarded forecasts remain documented in the relevant sections and the version history.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

References

- [1] S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. **61**, 1–23 (1989). [doi:10.1103/RevModPhys.61.1](https://doi.org/10.1103/RevModPhys.61.1).
- [2] Ya. B. Zeldovich, *The cosmological constant and the theory of elementary particles*, Sov. Phys. Usp. **11**, 381–393 (1968). [doi:10.1070/PU1968v011n03ABEH003927](https://doi.org/10.1070/PU1968v011n03ABEH003927).
- [3] P. J. E. Peebles and B. Ratra, *The cosmological constant and dark energy*, Rev. Mod. Phys. **75**, 559–606 (2003). [doi:10.1103/RevModPhys.75.559](https://doi.org/10.1103/RevModPhys.75.559).
- [4] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020). [doi:10.1051/0004-6361/201833910](https://doi.org/10.1051/0004-6361/201833910).
- [5] S. M. Carroll, *The cosmological constant*, Living Rev. Relativ. **4**, 1 (2001). [doi:10.12942/lrr-2001-1](https://doi.org/10.12942/lrr-2001-1).
- [6] A. G. Riess et al., *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, Astron. J. **116**, 1009–1038 (1998). [doi:10.1086/300499](https://doi.org/10.1086/300499).
- [7] S. Perlmutter et al., *Measurements of Ω and Λ from 42 high-redshift supernovae*, Astrophys. J. **517**, 565–586 (1999). [doi:10.1086/307221](https://doi.org/10.1086/307221).

- [8] L. M. Martyushev and V. D. Seleznev, *Maximum entropy production principle in physics, chemistry and biology*, Phys. Rep. **426**, 1–45 (2006).
[doi:10.1016/j.physrep.2005.12.001](https://doi.org/10.1016/j.physrep.2005.12.001).
- [9] A. A. Starobinsky, *A new type of isotropic cosmological models without singularity*, Phys. Lett. B **91**, 99–102 (1980). [doi:10.1016/0370-2693\(80\)90670-X](https://doi.org/10.1016/0370-2693(80)90670-X).
- [10] DESI Collaboration (M. Abdul-Karim et al.), *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, Phys. Rev. D **112**, 083515 (2025). [doi:10.1103/tr6y-kpc6](https://doi.org/10.1103/tr6y-kpc6). arXiv:2503.14738 [astro-ph.CO].
- [11] A. Eichhorn, *Asymptotically safe gravity*, arXiv:2003.00044 (2020).
- [12] A. Eichhorn and A. Held, *Top mass from asymptotic safety*, Phys. Lett. B **777**, 217–221 (2018). [doi:10.1016/j.physletb.2017.12.040](https://doi.org/10.1016/j.physletb.2017.12.040). arXiv:1707.01107.
- [13] W. Hu and I. Sawicki, *Models of $f(R)$ cosmic acceleration that evade solar-system tests*, Phys. Rev. D **76**, 064004 (2007). [doi:10.1103/PhysRevD.76.064004](https://doi.org/10.1103/PhysRevD.76.064004).
- [14] T. P. Sotiriou and V. Faraoni, *$f(R)$ theories of gravity*, Rev. Mod. Phys. **82**, 451–497 (2010). [doi:10.1103/RevModPhys.82.451](https://doi.org/10.1103/RevModPhys.82.451).
- [15] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge Monographs on Mathematical Physics, Cambridge University Press, 1982.
[doi:10.1017/CBO9780511622632](https://doi.org/10.1017/CBO9780511622632).
- [16] Ö. Sevinç, Ö. Ökcü, and E. Aydinler, *From MOND entropy to extended uncertainty principles: A unified framework*, Phys. Lett. B **873**, 140169 (2026).
[doi:10.1016/j.physletb.2026.140169](https://doi.org/10.1016/j.physletb.2026.140169). arXiv:2601.14353 [gr-qc].
- [17] C. Pfeifer, N. Voicu, A. Friedl–Szász, and E. Popovici-Popescu, *From kinetic gases to an exponentially expanding universe – the Finsler–Friedmann equation*, JCAP **2025**(10), 050 (2025). [doi:10.1088/1475-7516/2025/10/050](https://doi.org/10.1088/1475-7516/2025/10/050).
- [18] T. A. Hunt, *Dark Matter Phenomenology as Informational Persistence: Nonlocal Gravity, MOND, and the Survival of Curvature Under Coarse-Graining*, Zenodo preprint (2026). [doi:10.5281/zenodo.18215660](https://doi.org/10.5281/zenodo.18215660).
- [19] L. Geiger, *From Landscape to Proof: The Even-Dominance Route to the Riemann Hypothesis*, Zenodo preprint, 2026. [doi:10.5281/zenodo.19035640](https://doi.org/10.5281/zenodo.19035640).
- [20] L. Geiger, *FST Spectrum Duality: The Renormalized Free Energy Principle as Physical Instantiation of Functional Stability Theory*, Zenodo preprint, 2026. [doi:10.5281/zenodo.19036190](https://doi.org/10.5281/zenodo.19036190).
- [21] L. Geiger, *A Thermodynamic Obstruction to Finite-Time Blow-Up in the 3D Navier–Stokes Equations*, Zenodo preprint, 2026. [doi:10.5281/zenodo.19087449](https://doi.org/10.5281/zenodo.19087449).
- [22] L. Geiger, *Volume-Independent Spectral Gap for Strong-Coupling Lattice Gauge Theory via Poincaré–Dobrushin Machinery*, Zenodo preprint, 2026. [doi:10.5281/zenodo.19087433](https://doi.org/10.5281/zenodo.19087433).
- [23] L. Geiger, *The BSD Conjecture as a Positivity Normal-Form Theorem*, Zenodo preprint, 2026. [doi:10.5281/zenodo.19087443](https://doi.org/10.5281/zenodo.19087443).

- [24] E. J. Copeland, A. R. Liddle, and D. Wands, *Exponential potentials and cosmological scaling solutions*, Phys. Rev. D **57**, 4686–4690 (1998). doi:[10.1103/PhysRevD.57.4686](https://doi.org/10.1103/PhysRevD.57.4686).
- [25] M. Li, *A model of holographic dark energy*, Phys. Lett. B **603**, 1–5 (2004). doi:[10.1016/j.physletb.2004.10.014](https://doi.org/10.1016/j.physletb.2004.10.014).
- [26] R. Bousso and J. Polchinski, *Quantization of four-form fluxes and dynamical neutralization of the cosmological constant*, JHEP **0006**, 006 (2000). doi:[10.1088/1126-6708/2000/06/006](https://doi.org/10.1088/1126-6708/2000/06/006).
- [27] L. Susskind, *The anthropic landscape of string theory*, in: *Universe or Multiverse?*, Cambridge University Press, 247–266 (2007). doi:[10.1017/CBO9781107050990.018](https://doi.org/10.1017/CBO9781107050990.018).
- [28] N. Kaloper and A. Padilla, *Sequestering the standard model vacuum energy*, Phys. Rev. Lett. **112**, 091304 (2014). doi:[10.1103/PhysRevLett.112.091304](https://doi.org/10.1103/PhysRevLett.112.091304).
- [29] F. Lin and Y. Yang, *Existence of energy minimizers as stable knotted solitons in the Faddeev model*, Commun. Math. Phys. **249**, 273–303 (2004). doi:[10.1007/s00220-004-1110-y](https://doi.org/10.1007/s00220-004-1110-y).
- [30] C. Wetterich, *Dark energy evolution from quantum gravity*, arXiv:2407.03465 [hep-th] (2024).
- [31] D. Vasak, J. Kirsch, and J. Struckmeier, *Dark energy and inflation invoked in CCGG by locally contorted space-time*, Eur. Phys. J. Plus **135**, 404 (2020). doi:[10.1140/epjp/s13360-020-00415-7](https://doi.org/10.1140/epjp/s13360-020-00415-7).
- [32] I. L. Shapiro and J. Solà, *The scaling evolution of the cosmological constant*, JHEP **2002**(02), 006 (2002). doi:[10.1088/1126-6708/2002/02/006](https://doi.org/10.1088/1126-6708/2002/02/006). arXiv:hep-th/0012227.
- [33] J. Solà Peracaula, *The cosmological constant problem and running vacuum in the expanding universe*, Phil. Trans. R. Soc. A **380**, 20210182 (2022). doi:[10.1098/rsta.2021.0182](https://doi.org/10.1098/rsta.2021.0182).
- [34] A. G. Cohen, D. B. Kaplan, and A. E. Nelson, *Effective Field Theory, Black Holes, and the Cosmological Constant*, Phys. Rev. Lett. **82**, 4971 (1999). doi:[10.1103/PhysRevLett.82.4971](https://doi.org/10.1103/PhysRevLett.82.4971).
- [35] L. Geiger, *Functional Stability Theory – A Programmatic Hub: Universal Convexity-Uniqueness Across Scales (From Particles to Cosmology)*, Zenodo (2026). doi:[10.5281/zenodo.20130499](https://doi.org/10.5281/zenodo.20130499).
- [36] D. Efstratiou, E. A. Paraskevas, and L. Perivolaropoulos, *Addressing the DESI DR2 Phantom-Crossing Anomaly and Enhanced H_0 Tension with Reconstructed Scalar-Tensor Gravity*, Phys. Rev. D **113**, 123525 (2026). doi:[10.1103/zdcdg-4sdf](https://doi.org/10.1103/zdcdg-4sdf). arXiv:2511.04610 [astro-ph.CO].
- [37] S. Antusch, S. F. King, and X. Wang, *Coupled Dark Energy and Dark Matter for DESI: An Effective Guide to the Phantom Divide*, arXiv:2604.08449 [astro-ph.CO] (2026).

- [38] P. Thanankullaphong, P. Sahoo, P. H. Puttasiddappa, and N. Roy, *Quintom Dark Energy: Future Attractor and Phantom Crossing in Light of DESI DR2 Observations*, Phys. Rev. D **113**, 084069 (2026). doi:[10.1103/4b1x-6pcx](https://doi.org/10.1103/4b1x-6pcx). arXiv:2601.02284 [astro-ph.CO].